

BCA-5003: COMPUTER NETWORK

UNIT IV — COMPLETE DETAILED NOTES + TOPIC-WISE PYQ ANSWER BOOK

Network Devices • Bridges • Switching Devices • NIC • Firewall • UDP/TCP • Framing

Unit-I format: Index → detailed concepts → labelled diagrams/tables → topic-wise PYQs with years
→ long answers → revision plan

Prepared from the supplied Unit-IV notes. Raw-text inaccuracies have been corrected for accurate, exam-safe answers.

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No.	Topic	PYQ years found in supplied Unit-IV material
1	Unit-IV roadmap and answer-writing strategy	—
2	Network components and connecting devices	2017, 2018, 2023–24, 2025–26
3	Hub, switch, bridge, router, repeater, gateway, modem and access point	2019, 2021, 2022–23, 2023–24, 2025–26
4	Bridge: learning, filtering, forwarding and STP	2025–26
5	Hub versus switch; router versus bridge	2024–25
6	Modem, gateway, repeater, firewall and NIC	2019, 2022–23
7	UDP and TCP: headers, fields, reliability and sizes	2015, 2019
8	Character stuffing and bit stuffing	2017
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How to prepare this Unit

1. Memorize each network device by OSI layer, address type used, main job, and a real example.
2. Use the comparison tables for Hub vs Switch, Router vs Bridge, and TCP vs UDP.
3. Practise the bridge-learning and forwarding diagram, plus bit/character stuffing examples.
4. For header questions, draw neat field diagrams and write the exact minimum/maximum size.

1. Unit-IV Roadmap

Unit IV covers the devices that connect networks and the protocols/functions that support local delivery and transport. It is highly scoring because questions are often direct: “Explain bridge”, “Compare hub and switch”, “Write short note on firewall/NIC”, “TCP vs UDP”, and “Explain bit stuffing.”

Best answer format

For a network-device answer, follow this order: definition → OSI layer → working diagram → key functions → advantages/limitations → example. For a comparison, use a table instead of long paragraphs. For TCP/UDP headers, draw fields and state their sizes exactly.

2. Network Components and Connecting Devices

Network components are hardware and software elements used to connect devices, transfer data, direct traffic, provide network access, and deliver services. Important hardware components include network interface cards, cables, hubs, switches, bridges, routers, repeaters, modems, gateways, wireless access points, firewalls, and servers.

Typical small network diagram

Internet
■
[ISP modem / ONT] ■■ [Router + Firewall] ■■ [Switch] ■■ PC 1
■ ■■■ PC 2
■ ■■■ Server / Printer
■■■ [Wi-Fi Access Point]))) Laptop / Phone

Device	Primary job	Common OSI layer(s)
NIC	Gives a host a network interface; sends/receives frames/signals.	Physical + Data Link
Repeater	Regenerates/re-times a signal to extend a physical link.	Physical
Hub	Multiport repeater; copies incoming bits to other ports.	Physical
Bridge	Filters/forwards frames between LAN segments using MAC table.	Data Link
Switch	Multiport bridge; forwards Ethernet frames using MAC addresses.	Data Link
Router	Forwards IP packets between separate networks.	Network
Gateway	Connects systems/protocols with different architectures; meaning depends on context.	Often higher layers; may be multi-layer
Modem	Modulates/demodulates where required by an access technology.	Physical
Firewall	Enforces security policy by permitting/denying traffic.	Can operate at multiple layers

3. Important Network Devices

3.1 Network Interface Card (NIC)

A Network Interface Card, also called a network adapter or network interface controller, is the hardware that enables a computer or other host to connect to a network. It may be an Ethernet adapter, Wi-Fi adapter, built-in motherboard interface, or USB adapter. The NIC performs Physical-Layer signal functions and Data-Link-Layer operations such as MAC addressing and frame handling.

NIC diagram

[Computer / OS] ■ [NIC: MAC address + Ethernet/Wi-Fi hardware] ■ Cable / Radio medium ■ Network

3.2 Repeater

A repeater is a Physical-Layer device that receives a weakened or distorted signal, regenerates/re-times it, and transmits a clean version onward to extend distance. A digital repeater is not merely an amplifier: it interprets/regenerates bits and signals rather than just increasing power. It does not read MAC/IP addresses and does not make forwarding decisions.

Repeater diagram

Long cable segment ■■ weak/distorted signal ■■■ [REPEATER] ■■ regenerated signal ■■■ next segment

3.3 Hub

A hub is essentially a multiport repeater. When a signal arrives on one port, the hub repeats it to all other ports. It does not learn MAC addresses or selectively forward frames. All attached devices share the same collision domain in traditional half-duplex Ethernet, which reduces performance as traffic grows. Hubs are now largely obsolete.

Hub behaviour

PC A ■■
PC B ■■■■ [HUB] ■■ Incoming bits from A copied to B, C, D ...
PC C ■■
PC D ■■

3.4 Switch

An Ethernet switch is a multiport Layer-2 bridge. It learns source MAC addresses and stores them in a MAC/CAM table. When a frame arrives, it checks the destination MAC and forwards the frame only to the correct port if known. If destination is unknown or broadcast, it floods the frame within the VLAN except the incoming port. Each switch port is a separate collision domain; modern switched Ethernet generally uses full-duplex operation.

Switch MAC learning / forwarding diagram

PC A (MAC A) ■ Port 1 ■
PC B (MAC B) ■ Port 2 ■■■ [SWITCH: MAC table A→1, B→2, C→3] ■■ Port 3 ■ PC C (MAC C)
PC A → PC C: switch receives on Port 1, finds C→3, forwards only Port 3.

3.5 Router

A router is a Layer-3 device that connects different IP networks and forwards packets based on destination IP address and a routing table. Routers separate broadcast domains. A home Wi-Fi router often combines router, switch, wireless access point, NAT, DHCP server, and firewall functions in one box.

Router diagram

LAN 192.168.1.0/24 ■■ [ROUTER: routing table] ■■ WAN / Internet ■■ Remote network 10.0.0.0/24

3.6 Gateway

A gateway connects networks or systems that may use different protocols, data formats, or architectures. In everyday IP networking, “default gateway” usually means the router that sends a host’s off-subnet traffic to other networks. In broader networking, an application gateway may translate between dissimilar protocols or services.

3.7 Modem

Modem means modulator-demodulator. Historically, it converted a computer’s digital data to analog signals for telephone lines and converted received analog signals back to digital form. In modern broadband, the exact physical conversion depends on technology: DSL modem, cable modem, cellular modem, or fibre ONT. A modem is distinct from a router, although consumer devices often combine both.

3.8 Server

A server is a computer or program that receives requests and provides services/resources to clients over a network. Examples include web servers, file servers, database servers, DNS servers, mail servers, proxy servers, and game servers.

4. Bridge: Detailed Study

A bridge is a Data-Link-Layer device that connects two or more LAN segments and selectively forwards or filters Ethernet frames according to MAC addresses. A switch is essentially a high-port-density, hardware-accelerated bridge. Bridges reduce unnecessary traffic crossing between segments, but they do not normally separate broadcast domains unless VLAN/routing functions are involved.

Bridge diagram

LAN Segment 1 LAN Segment 2
PC A ■■ PC B ■■ [BRIDGE] ■■ PC C ■■ PC D ■■
■ reads source/destination MAC addresses ■

How a transparent bridge works

1. Learning: on receipt of a frame, it learns the source MAC address and associates it with the incoming port.
2. Lookup: it checks the destination MAC address in its forwarding/bridge table.
3. Filtering: if destination is known on the same incoming port/segment, it does not forward the frame.
4. Forwarding: if destination is known on another port, it forwards only to that port.
5. Flooding: if destination is unknown, or the frame is broadcast, it forwards/floods it to relevant other ports.

Bridge forwarding example

Bridge table: A → left port, B → left port, C → right port, D → right port.
Frame A → C arrives from left. Bridge learns A on left, finds C on right, forwards only right.
Frame A → B arrives from left. Bridge learns A on left, finds B also left, filters it—does not cross bridge.
Frame A → unknown E. Bridge floods right because destination is not known.

Bridge functions

Function	Explanation
Segmentation	Divides a larger LAN into separate collision domains, reducing local collisions/traffic.
Learning	Builds MAC table dynamically from source addresses of arriving frames.
Filtering	Keeps local traffic in its local segment when destination is also local.
Forwarding	Sends frame to the appropriate segment when destination is on another segment.
Loop avoidance	Uses Spanning Tree Protocol (STP) in bridged Ethernet to block redundant paths and prevent loops/broadcast storms.
Interconnection	Can connect compatible LAN segments; special translation/encapsulation bridges historically connected differing link technologies.

Important correction

A bridge does not normally “forward packets despite faulty nodes” and does not route using IP addresses. It forwards frames based on MAC addresses. It also does not eliminate broadcasts; unknown-unicast and broadcast frames are typically flooded within the bridged broadcast domain.

5. Device Comparisons

Hub versus Switch

Basis	Hub	Switch
OSI layer	Physical Layer (Layer 1).	Data-Link Layer (Layer 2) in a normal Ethernet switch.
Forwarding method	Repeats incoming bits to all ports.	Learns MAC addresses and selectively forwards frames.
Collision domain	One shared collision domain for all ports.	One collision domain per port.
Duplex	Typically half-duplex.	Usually full-duplex on modern Ethernet links.
Bandwidth	Shared among connected devices.	Dedicated link bandwidth per switched port.
Security/performance	Low; all traffic copied to all ports.	Better; unicast goes to selected port when MAC known.
Modern use	Mostly obsolete.	Standard LAN device.

Router versus Bridge

Basis	Router	Bridge
OSI layer	Network Layer (Layer 3).	Data-Link Layer (Layer 2).
Address used	IP address.	MAC address.
Unit forwarded	Packet.	Frame.
Purpose	Connects separate IP networks.	Connects/segments LANs at Layer 2.
Table used	Routing table.	MAC/forwarding/bridge table.
Broadcast domains	Separates broadcast domains.	Normally remains within same broadcast domain.
Decision	Chooses next hop/path based on IP routes.	Forwards/filters based on destination MAC location.

Basis	TCP	UDP
Connection	Connection-oriented.	Connectionless.
Service	Reliable ordered byte stream.	Best-effort datagrams; no built-in reliability/order.
Reliability tools	Sequence numbers, ACKs, retransmission, checksum, timers.	Checksum for corruption detection; no ACK/retransmission/order mechanism in U
Flow/congestion control	Yes, provided by TCP.	No built-in transport flow/congestion control.
Header	20–60 bytes.	Fixed 8 bytes.
Overhead	Higher.	Lower.
Examples	HTTP(S), SSH, SMTP, IMAP, file transfers.	DNS queries, DHCP, voice/video, gaming, streaming protocols where app may ha

TCP fields absent from UDP and why

UDP does not contain TCP's sequence number, acknowledgement number, data offset/options, control flags, window size, and urgent pointer. These are absent because UDP does not establish a connection and does not itself provide ordered delivery, acknowledgement/retransmission, flow control, congestion control, or advanced connection state.

Datagram sizes

UDP length field is 16 bits and includes header plus data. Therefore the theoretical maximum UDP datagram length is 65,535 bytes, including the 8-byte header; maximum UDP payload is 65,527 bytes. In practice, IP path MTU and fragmentation constraints often limit useful UDP payloads to much smaller values. TCP has no equivalent single "maximum application message" dictated by a 16-bit UDP-style length field; individual TCP segments are constrained by MSS/path MTU, while TCP presents a byte stream to applications.

8. Character Stuffing and Bit Stuffing

Framing requires a receiver to recognize the beginning and end of each frame. If a special delimiter appears inside user data, the receiver might mistake it for a frame boundary. Stuffing methods solve this ambiguity.

Character / byte stuffing

Character stuffing is used in byte-oriented protocols. A special flag byte marks frame boundaries. If the flag byte—or escape byte—appears in the payload, the sender inserts an escape byte before it. The receiver removes the escape byte during destuffing.

Example: let FLAG = F and ESC = E

```
Original data: A F B E C
Stuffed payload: A E F B E E C
Transmitted frame: F A E F B E E C F
Receiver removes added E values and restores A F B E C.
```

Bit stuffing

Bit stuffing is used in bit-oriented protocols such as HDLC. A flag pattern, commonly `01111110`, marks the start and end of a frame. To prevent this exact flag pattern from appearing inside data, the sender inserts a `0` after every sequence of five consecutive `1` bits in the data. The receiver removes each such inserted 0 after five consecutive 1s.

Bit-stuffing example

```
Data bits: 01111110 111110 001
After stuffing: 01111110 1111100 001
Frame transmitted: 01111110 | stuffed data bits | 01111110
Receiver destuffs by removing the 0 that follows every five consecutive 1s in payload.
```

Character stuffing versus bit stuffing

Basis	Character stuffing	Bit stuffing
Unit handled	Bytes/characters.	Bits.
Frame delimiter	Special flag byte/character.	Special bit flag, commonly 01111110.
Escape action	Insert escape byte before reserved byte.	Insert 0 after five consecutive 1s in data.
Used by	Byte-oriented framing methods.	Bit-oriented protocols such as HDLC.

9. Topic-Wise PYQ Answer Bank

Each answer is written in long, exam-ready form. Use the diagrams/tables from the relevant concept section while practising in your notebook.

Q1. Explain the use/functions of bridges.

Asked: 2025–26

Answer

A bridge is a Data-Link-Layer device used to connect two or more LAN segments. It examines Ethernet frame MAC addresses and decides whether to forward the frame to another segment or filter it locally. It is more intelligent than a repeater because it makes forwarding decisions at Layer 2.

- 1. Segmentation:** A bridge divides a large LAN into smaller collision domains. This reduces collisions and unnecessary local traffic, improving performance.
- 2. Learning:** A bridge learns MAC addresses dynamically. It records the source MAC address of each incoming frame together with the port on which that frame arrived. This creates a bridge/MAC forwarding table.
- 3. Filtering:** If the destination MAC address is on the same segment as the sender, the bridge filters the frame and does not transmit it to the other segment. This prevents unnecessary traffic crossing the bridge.
- 4. Selective forwarding:** If the destination is on another segment, the bridge forwards the frame only to that segment instead of sending it everywhere.
- 5. Flooding when destination is unknown:** If the destination MAC is not in the bridge table, the bridge floods the frame to relevant other ports. Broadcast frames are also normally flooded within the same broadcast domain.
- 6. Loop avoidance:** In networks with redundant bridge paths, Spanning Tree Protocol prevents Layer-2 loops and broadcast storms by blocking redundant paths.

Conclusion: Bridges improve LAN efficiency through MAC learning, filtering, forwarding, and segmentation. They forward frames, not IP-routed packets.

Q2. What are network components? Explain.

Asked: 2023–24, 2025–26

Answer

Network components are the hardware and software elements used to establish communication, connect devices, transfer data, control traffic, provide access to networks, and deliver network services.

NIC: connects an individual host to wired Ethernet or Wi-Fi and provides MAC/frame-level capabilities.

Cables/connectors: carry signals in wired networks, such as twisted pair, coaxial, and fibre; connectors terminate/attach cables to devices.

Hub: a multiport Physical-Layer repeater that copies bits to all ports; mostly obsolete.

Switch: a Layer-2 multiport bridge that learns MAC addresses and forwards frames selectively.

Bridge: connects LAN segments and filters/forwards frames using MAC addresses.

Router: connects distinct IP networks and forwards packets using destination IP address and routing table.

Modem/ONT: adapts signals for a provider access technology, for example DSL, cable, cellular, or fibre access.

Gateway: provides an exit/translation point between networks or protocols; default gateway usually refers to the local router.

Firewall: protects the network using traffic filtering rules and security inspection.

Servers: provide services such as web, file, database, DNS, email, authentication, and applications to clients.

Conclusion: Together, these components provide connectivity, local delivery, routing, security, and services in a functioning network.

Q3. Compare hub and switch; compare router and bridge.

Asked: 2024–25

Answer

Hub vs Switch: A hub works at the Physical Layer and repeats incoming bits to all ports. It does not examine MAC addresses, all ports share one collision domain, and it normally uses half-duplex communication. A switch works mainly at the Data-Link Layer, learns MAC addresses, forwards unicast frames only to the selected destination port, creates a separate collision domain per port, and normally supports full-duplex communication. Switches provide much better performance and security than hubs.

Router vs Bridge: A router works at the Network Layer and forwards IP packets between separate IP networks using routing tables. It separates broadcast domains. A bridge works at the Data-Link Layer and forwards/filters MAC frames between LAN segments using a bridge table. It normally does not break the Layer-2 broadcast domain. Therefore, a router uses IP addresses and routing decisions; a bridge uses MAC addresses and local forwarding decisions.

Q4. Write a short note on modem, gateway, repeater, firewall, or NIC.

Asked: 2019, 2022–23

Answer

Modem: A modulator-demodulator adapts digital data to signals suitable for an access medium and recovers digital data at the receiving side. Traditional modems converted digital data to analog telephone-line signals; modern variants include DSL/cable/cellular modems and fibre ONTs.

Gateway: A gateway connects a local network to another network or translates between systems/protocols. In normal IP terminology, the default gateway is the router used to send traffic outside a local subnet.

Repeater: A Physical-Layer device that regenerates/re-times weakened signals to extend link distance. It does not examine MAC or IP addresses and does not make routing decisions.

Firewall: A security device/software that allows, blocks, logs, and inspects traffic according to security rules. It protects internal systems from unauthorised network traffic.

NIC: A hardware network adapter that allows a host to connect to Ethernet/Wi-Fi. It performs physical signalling and Layer-2 frame/MAC functions.

Q5. What is the maximum size of a UDP datagram? Compare TCP and UDP headers. List TCP fields absent from UDP, with reason.

Asked: 2015, 2019

Answer

UDP has a fixed 8-byte header containing source port, destination port, length, and checksum. Its 16-bit length field represents the complete UDP datagram including header and payload. Therefore maximum UDP datagram size is **65,535 bytes**. The maximum UDP payload is $65,535 - 8 = 65,527$ bytes. Actual useful payload may be smaller because of IP MTU and fragmentation limits.

TCP has a minimum 20-byte header and maximum 60-byte header. TCP header includes source/destination ports, sequence number, acknowledgement number, data offset, flags, window size, checksum, urgent pointer, and optional options/padding. UDP has no sequence number, acknowledgement number, data offset/options, flags, window field, or urgent pointer.

These TCP fields are absent from UDP because UDP is connectionless and intentionally lightweight. UDP does not provide a connection state, ordered byte stream, acknowledgement/retransmission, flow control, congestion control, or urgent-data mechanism. TCP provides these features and therefore needs the extra header fields.

Conclusion: UDP is selected for low overhead and time-sensitive applications where loss can be tolerated or handled by the application; TCP is selected where reliable, ordered delivery is required.

Q6. Explain character stuffing and bit stuffing.

Asked: 2017

Answer

Character stuffing and bit stuffing are framing methods used to ensure that reserved frame-boundary delimiters are not mistaken for normal data.

Character stuffing: A byte-oriented protocol uses a special flag character/byte to mark start and end of a frame. If the flag itself appears in payload data, the sender inserts an escape character before it. If the escape character appears in payload, it is also

escaped. The receiver removes inserted escape characters before delivering data to upper layer.

Example: If FLAG = F and ESC = E, payload `A F B E C` becomes `A E F B E E C` inside the frame. The transmitted data is `F | A E F B E E C | F`.

Bit stuffing: A bit-oriented protocol such as HDLC uses flag pattern `01111110`. To prevent this pattern from appearing in payload, sender inserts a 0 after every occurrence of five consecutive 1s in data. Receiver removes that 0 during destuffing.

Example: Whenever payload contains `11111`, sender transmits `111110`. This prevents six consecutive 1s from creating the flag pattern in normal data.

Conclusion: Character stuffing works with bytes and escape characters; bit stuffing works with bits and inserts zeros after five 1s. Both preserve unambiguous frame boundaries.

10. Final Revision Sheet

Most important diagrams to practise

- Complete home/office network: modem/ONT → router/firewall → switch → PCs/server/Wi-Fi.
- Hub broadcasting to all ports versus switch selective forwarding using MAC table.
- Bridge connecting two LAN segments with MAC learning/filtering/forwarding.
- Router connecting two IP networks.
- Firewall between Internet and internal LAN.
- UDP 8-byte header and TCP 20–60 byte header.
- Character-stuffing and bit-stuffing examples.

One-line revision table

Term	Remember this
Hub	Layer 1 multiport repeater; repeats bits to all ports.
Switch	Layer 2 multiport bridge; MAC-based forwarding; one collision domain per port.
Bridge	Layer 2 device that learns MACs and filters/forwards frames between segments.
Router	Layer 3 device that forwards IP packets between networks and separates broadcast domains.
Repeater	Layer 1 signal regeneration; no MAC/IP decision.
Firewall	Security policy enforcement for traffic; may work at L3/L4/L7.
NIC	Host interface with physical and data-link functions.
UDP	Connectionless 8-byte header; max datagram 65,535 bytes.
TCP	Reliable ordered transport; 20–60 byte header.
Byte stuffing	Escape reserved delimiter bytes in byte-oriented framing.
Bit stuffing	Insert 0 after five 1s in bit-oriented framing.

7-day Unit-IV plan

Day	Target	Output
1	All device definitions + OSI layers	Make flashcards for NIC/repeater/hub/switch/bridge/router/gateway/modem.
2	Bridge and switch working	Draw bridge MAC learning and switch forwarding diagrams; write Q1.
3	Comparison practice	Write Hub vs Switch and Router vs Bridge tables from memory.
4	Firewall, NIC, modem/gateway/repeater	Write Q4 in your notebook; practise firewall diagram.
5	TCP and UDP	Draw headers; learn exact sizes and missing TCP fields; write Q5.
6	Stuffing/framing	Practise character and bit stuffing; write Q6.
7	PYQ mock test	Write Q1–Q6 under time limit; redraw all key diagrams.

Accuracy reminders

- A switch does flood unknown-unicast and broadcast frames when required; it does not always send every frame only to one port.
- A bridge forwards Layer-2 frames based on MAC addresses; a router forwards Layer-3 packets based on IP addresses.
- A repeater regenerates a signal rather than performing address-aware forwarding.
- UDP maximum datagram is 65,535 bytes including its 8-byte header; its maximum payload is 65,527 bytes.
- TCP minimum header is 20 bytes and maximum header is 60 bytes.
- Bit stuffing inserts 0 after five consecutive 1s in payload; it is associated with bit-oriented protocols such as HDLC.

For 10-CGPA-level preparation: redraw diagrams without seeing the PDF, make the comparison tables from memory, and write all PYQs by hand at least twice.